1. Suppose that it takes at least 9 votes from a 12-member jury to convict a defendant. Suppose also that the probability that a juror votes a guilty person innocent is .2, whereas the probability that the juror votes an innocent person guilty is .1. If each juror acts independently and if 65 percent of the defendants are guilty, find the probability that the jury renders a correct decision. What percentage of defendants is convicted?

2. Suppose that, in flight, airplane engines will fail with probability $1 - p$, independently from engine to engine. If an airplane needs a majority of its engines operative to complete a successful flight, for what values of p is a 5-engine plane preferable to a 3-engine plane?

3. Approximately $80,000$ marriages took place in the state of New York last year. Estimate the probability that, for at least one of these couples,

   (a) both partners were born on April 30;

   (b) both partners celebrated their birthday on the same day of the year.

   State your assumptions.

4. How many people are needed so that the probability that at least one of them has the same birthday as you is greater than $\frac{1}{2}$

5. Suppose X is a discrete random variable taking values on $\{1, 2, 3, \dots\}$ and the distribution of X has the memory-less property :

$$P(X > s + t | X > s) = P(X > t)$$

   (a) Show that $P(X > 2) = P(X > 1)^2$

   (b) Using induction, show that $P(X > n) = P(X > 1)^n$

   (c) Show that X follows a Geometric distribution.

6. Imagine a lake containing an unknown total population of N fish. You capture K fish, tag them, and release them back into the water. Later, you capture a second sample of n fish, observing that exactly k of them possess tags.

   (a) If your second sample consisted of just a single fish, what is the likelihood that it is tagged?

   (b) Assuming your second sample size is $n > 1$, provide an estimate for the total population N based on the given information.

   (c) Let X represent the random variable for the number of tagged fish in your second sample. Determine the probability $P(X = k)$.

   (d) Calculate the expected value, $E(X)$, for the number of tagged fish caught.